

Ashwin Rachha

Backend Engineer | AI Platforms & Financial Infrastructure

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Builds APIs, banking integrations, and asynchronous workflows for AI and fintech products. Experience includes tenant-scoped authorization, secure token management, underwriting, and ML serving.

EXPERIENCE

Loan Labs | Applied AI Engineer | Jan 2026–Present

- Built agent-facing Rails APIs, borrower email intake, and an in-product conversational interface so users could handle loan documents and initiate agent actions in LoanOS.
- Designed fail-closed authorization for Composio integrations: tenant/owner scoping, separate write/send/merge/archive permissions, reviewed tool allowlists, and execution-time checks that invalidate revoked access.
- Re-architected Lois from one-off Ruby LLM calls into a LangGraph agentic system on Amazon Bedrock AgentCore for mortgage-document classification, lender-specific renaming, and policy validation.

Finally | AI Product Engineer / Tech Lead | Feb 2024–Jan 2026

- Architected reusable Plaid/Teller infrastructure for account linking, encrypted token storage, token lifecycle management, webhooks, transaction synchronization, and normalized account data.
- Built cash-based underwriting with 90-day bank data, daily-balance reconstruction, weekly recalculation, audit history, and manual overrides; supported \$3M+ in credit for 50+ companies in approximately three months.
- Joined as the first AI Product Engineer and led a three-engineer team from prototype to production on Classify AI; processed 50K+ transactions daily and reduced manual categorization by approximately 80%.
- Built retrieval-augmented classification with LangChain, Pinecone, Elasticsearch, Redis, Celery, and Django, combining transaction history, merchant enrichment, and custom charts of accounts.

UNAR Labs | Machine Learning Engineer | Jun 2023–Aug 2023

- Built accessibility-focused backend and data pipelines for visually impaired users with OpenCV, PyTorch, Transformers, and FastAPI; deployed on GCP with Docker.

Outreach | Machine Learning Platform Engineer Intern | May 2022–Aug 2022

- Built reusable NLP inference and deployment infrastructure using PySpark, MLflow, ONNX, NVIDIA Triton, Go/Python microservices, Docker, CI/CD, and GKE.

SKILLS

Python, Go, TypeScript, SQL · Django, FastAPI, Rails APIs · PostgreSQL, Redis, Celery, Elasticsearch · Plaid, Teller, webhooks, OAuth · AWS, GCP, Docker, Kubernetes, GKE · LangGraph, Bedrock AgentCore, MLflow, ONNX, Triton

EDUCATION & RESEARCH

Virginia Tech | M.S. Computer Science (Thesis) | 2021–2023 | GPA 4.0/4.0

PICT | B.E. Computer Science | 2016–2020 | GPA 8.70/10.00

Gurukul · LLM-enhanced CS education — Built an adaptive learning environment using RAG and guardrails; research at IEEE FIE 2024 and IEEE SouthEastCon 2023. [Thesis](#) · [Code](#)